Formal Disk4 – conditional filter pipeline

source: current `FILTER_SPECS` and runner control flow

Production acceptance sequence, with quotient exits, proof-producing rejections, incomplete exits, and the numerical geometry boundary.

1. Structure candidate

Contact graph / planar map
tile count and boundary incidence
optional imposed rotational symmetry
catalogue case or Graph Lab proposal

2. Mapping proposal

Direct or reflected copy orientation
cyclic contour offset per copy
interface ports and contour sequences
No weak order fixed yet

3. Assignment gates

`assignment.canonical`
fail → equivalent assignment

`assignment.cyclic_shift_compatible`
fail → outside imposed domain

4. Weak-order decision loop

Append one coincidence block, then run stable prefix gates.
Conventional, learned, and joint-LP proposal modes meet here.
Detailed on page 2.

5. Complete weak leaf

`weak.leaf_canonical`
`weak.complete_length`
`weak.complete_angle`
`weak.complete_topology.hard_outer_endpoint`
pass → emit `Placement`

6. Compile + pre-word topology

Compile interface words from the placement.

`preword.topology.hard_outer_endpoint`
`preword.topology.opposite_circular_sign`

fail → certified obstruction for this placement

7. Pre-word joint metric gate

Production decision: joint length / radius / smooth-turn / point-turn feasibility.

The eight `preword.metric.*` labels below are its instrumented curriculum, not eight independent production branches.

8. Exact word resolution

`word.exact_contradiction`
`word.unsupported_cycle`
`word.resource_limit`
`word.family_emitted`

Unsupported / budget exit ≠ impossibility

9. Family specialization + profile

Expand or rank supported word-family specializations.
Rebuild terminal contour variables.
Apply formal profile filters and exact deduplication.
`profile.emitted`

10. Geometry + assembly

Closure precheck
staged nonlinear optimization
contour validation
propagate copy isometries
validate interfaces and cycles

numerical failure is not a formal proof

Instrumented pre-word metric curriculum (ordered diagnostic vector)

`positive_lengths` → `local_radius_transport` → `local_smooth_turn` → `point_turn` → `global_radius_balance` → `global_turn_balance` → `strict_concavity` → `isoperimetric`

These counters expose where the combined pre-word system loses feasibility; the production verdict remains the combined gate in step 7.

Proof-producing rejection

A necessary condition is infeasible. This branch cannot yield a valid formal profile.

Quotient exit

An equivalent representative remains. No mathematical candidate is lost.

Incomplete exit

Timeout, resource limit, or unsupported word language. Preserve or defer the case.

Imposed-domain exit

The candidate violates the question deliberately asked by the selected symmetry / shard.

Numerical outcome

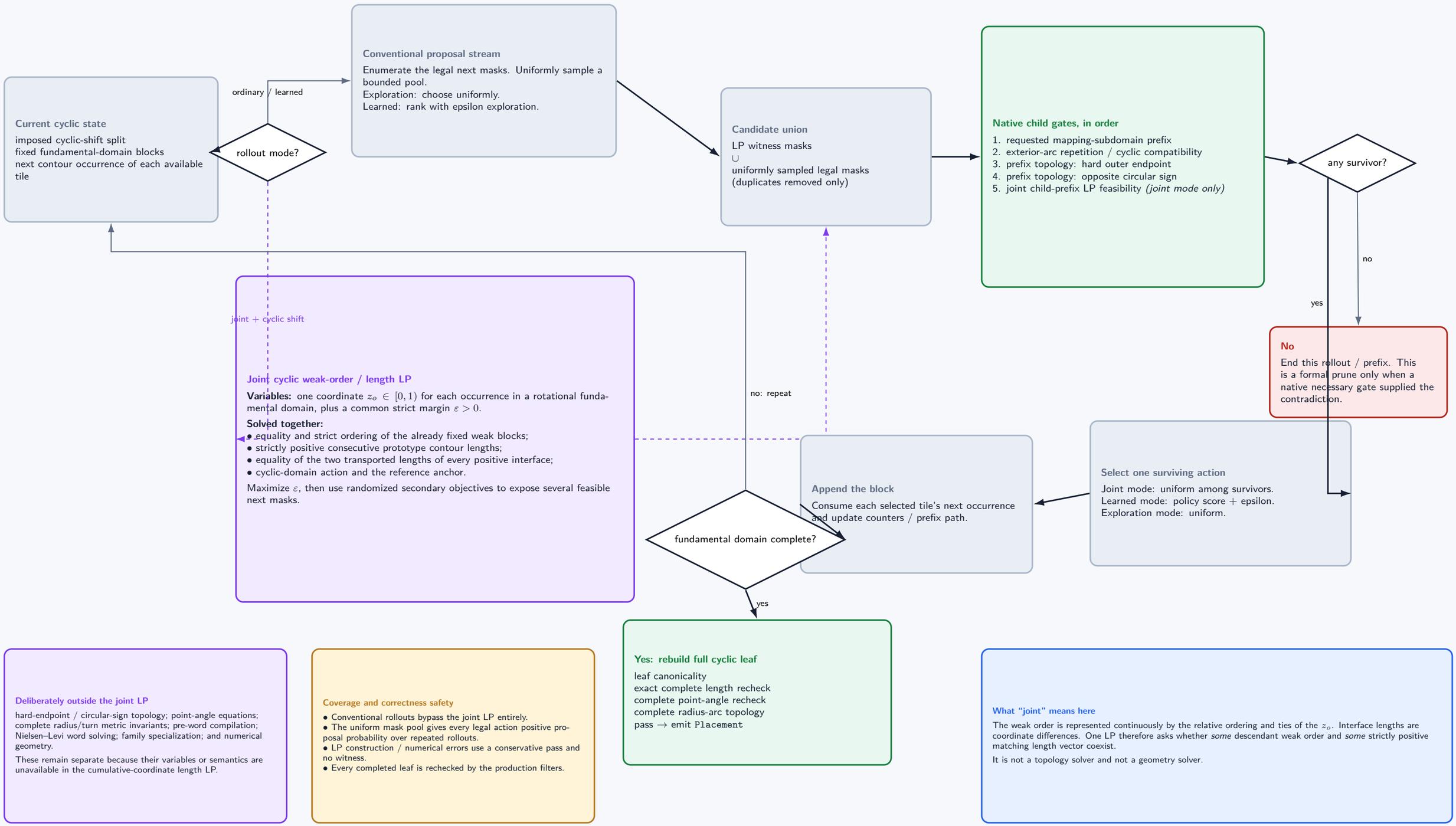
Found geometry is a witness. Failure to find one is not generally a non-existence proof.

Reading rule: every solid arrow means "the current candidate passed the preceding production gate." Side exits have deliberately different logical meanings. The joint LP shown on page 2 changes only the cyclic weak-order search route and never replaces the downstream exact checks.

Zoom: weak-order rollout and the joint length oracle

current configuration: 50% joint rollouts, 16 uniform masks + LP witnesses

The purple branch jointly solves a continuous weak-order extension and the interface-length equations; all other predicates stay in their native filters.



Logical contract: an LP witness influences proposal order, but it does not bypass a native filter. In joint mode, an infeasible child LP prunes that joint rollout; the separate conventional stream preserves global stochastic coverage against floating-oracle mistakes.